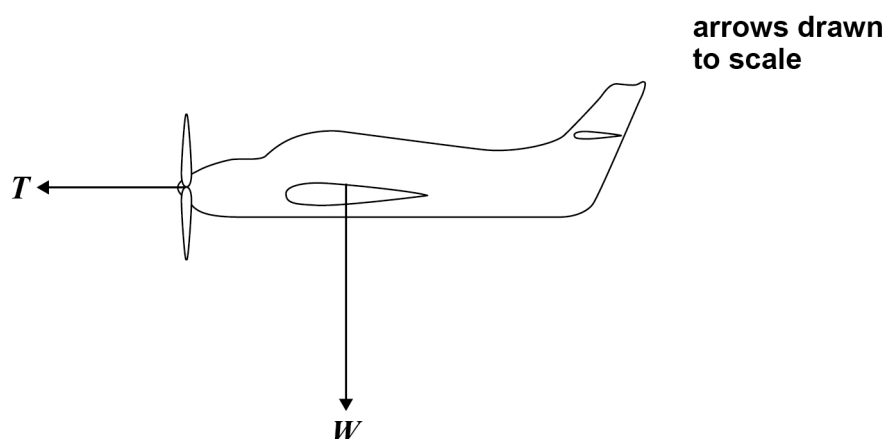


Figure 2 shows an aircraft that is descending. The horizontal and vertical components of velocity are constant. The thrust T and the weight W are shown. The lengths of the arrows indicate the magnitudes of the forces. There is no resultant moment on the aircraft.

Figure 2



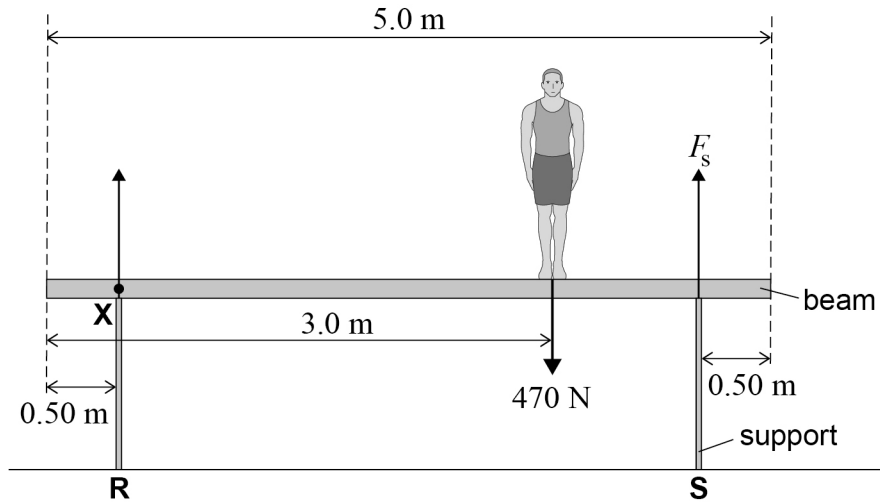
Draw and label on **Figure 2** arrows of suitable lengths to represent the lift L and the drag D that act on the aircraft.

[3 marks]

Don't write

Figure 2 shows a gymnast standing in equilibrium on a uniform beam, 3.0 m from one end.

Figure 2



The beam is 5.0 m long and the supports **R** and **S** are 0.50 m from each end. The weight of the beam is 700 N. The gymnast has a weight of 470 N.

F_s is the force exerted by **S** on the beam.

By taking moments about **X**, calculate the magnitude of F_s .

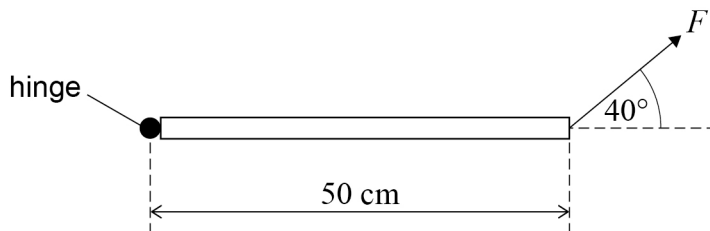
[3 marks]

magnitude of F_s = _____ N

Do not write

1 5

The diagram shows a hinged uniform bar of length 50 cm and weight 3.7 N.



The bar is held in a horizontal position by a force F that acts at 40° to the bar.

What is F ?

[1 mark]

A 2.4 N

☐

B 2.9 N

☐

C 4.9 N

☐

D 5.8 N

☐

Do not write	Do not write
--------------	--------------

0	3
---	---

A photon of electromagnetic radiation can be converted into an electron and a second particle in a pair-production event.

0	3	.	1
---	---	---	---

Identify the second particle that is created in this pair-production event.

[1 mark]

0	3	.	2
---	---	---	---

Explain why the photon energy must be greater than a minimum value for the pair-production event to happen.

[2 marks]



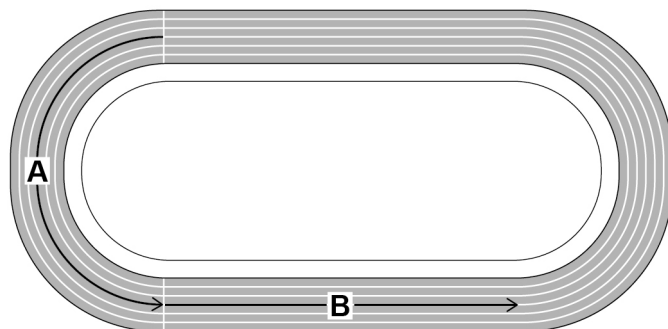
0 4 . 1

State what is meant by a vector quantity.

[1 mark]

Figure 1 shows an athletics track.

Figure 1



0 4 . 2

An athlete runs at constant speed through sections **A** and **B** of the track.

State and explain whether the athlete is accelerating in sections **A** and **B**.

[3 marks]

A _____

B _____



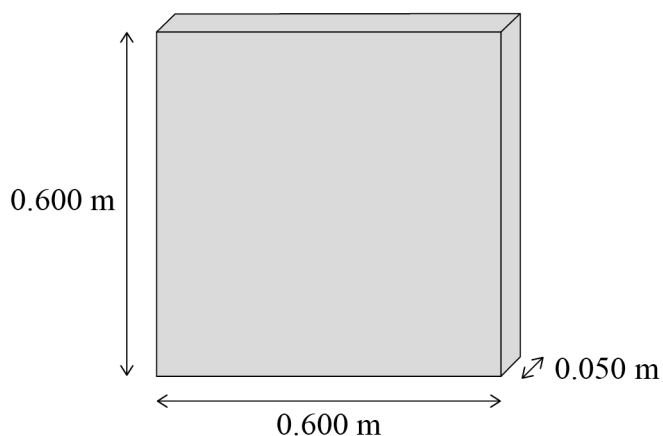
0 5 . 1

State what is meant by the centre of mass of an object.

[1 mark]

A uniform concrete block has the dimensions shown in **Figure 2**.

Figure 2



The weight of the concrete block is 343 N.

0 5 . 2

Calculate the density of the concrete.

[2 marks]

density = _____ kg m^{-3}

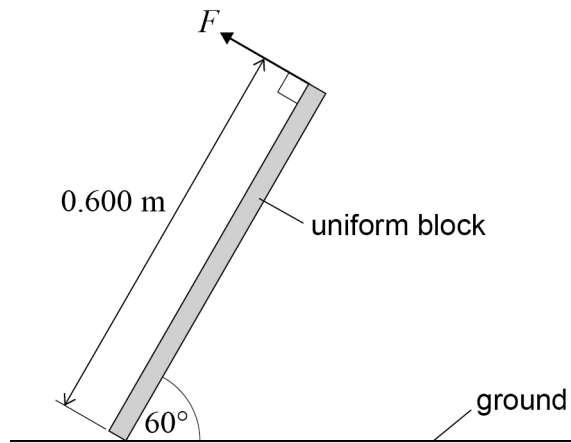


0 5 . 3

Figure 3 shows the uniform block held in equilibrium by a force F that is acting at 90° to the block.

Do not write
outside the
box

Figure 3



Calculate the magnitude of force F .

[3 marks]

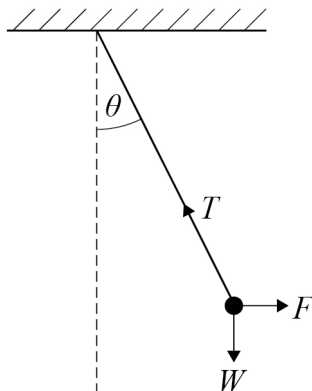
$F =$ _____ N

6

Turn over ►

Do not write

The bob of a pendulum is held in equilibrium as shown in the diagram.



Which expression is correct?

[1 mark]

A $F = T \times \cos \theta$

☐

B $F = T \times \sin \theta$

☐

C $\frac{\sin \theta}{F} = T$

☐

D $\frac{F}{\cos \theta} = T + W$

☐

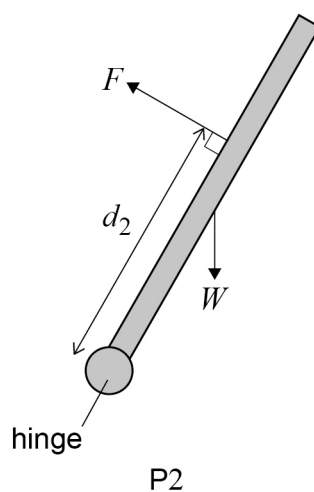
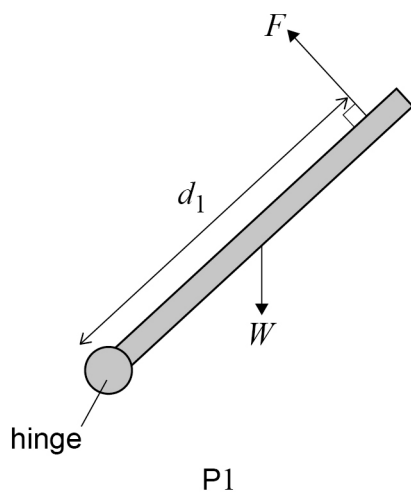
Turn over for the next question



Do not write Do not write

1 4

A hinged pole of weight W is in equilibrium at positions P1 and P2. Force F has the same magnitude in both positions. F is applied at right angles to the pole at different distances d_1 and d_2 from the hinge.



Which row is correct?

[1 mark]

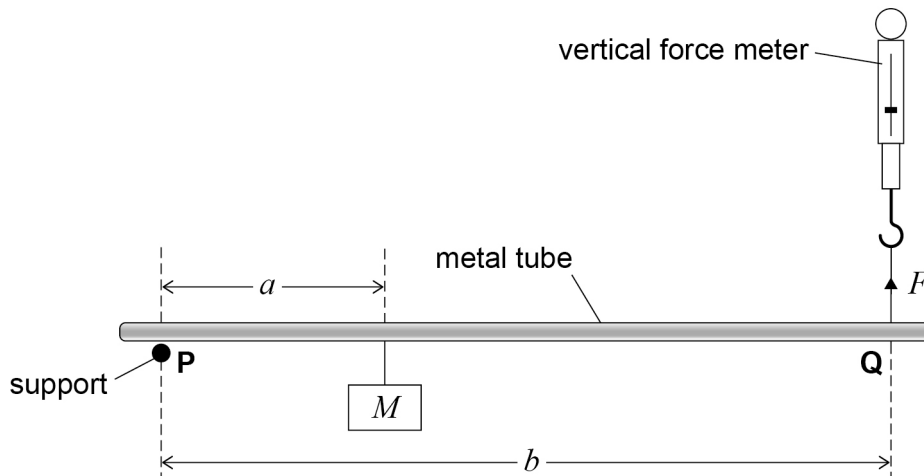
	Clockwise moment about the hinge	Anticlockwise moment about the hinge	
A	larger in P1 than in P2	smaller in P1 than in P2	☐
B	larger in P1 than in P2	larger in P1 than in P2	☐
C	smaller in P1 than in P2	smaller in P1 than in P2	☐
D	smaller in P1 than in P2	larger in P1 than in P2	☐



Don't write

The apparatus in **Figure 7** is used to determine the mass M of a load, using the principle of moments.

Figure 7



A uniform metal tube of weight w is kept horizontal by a support at **P** and by a vertical force F applied by a force meter at **Q**.

P and **Q** are the same distance from their respective ends of the metal tube.

A load of mass M is suspended from the metal tube at a distance a from **P**.

The distance between **P** and **Q** is b .

The distance a is varied over the range 0.100 m to 0.700 m.

The force F needed to keep the metal tube horizontal is measured for each value of a .

Throughout the experiment $b = 0.800$ m.

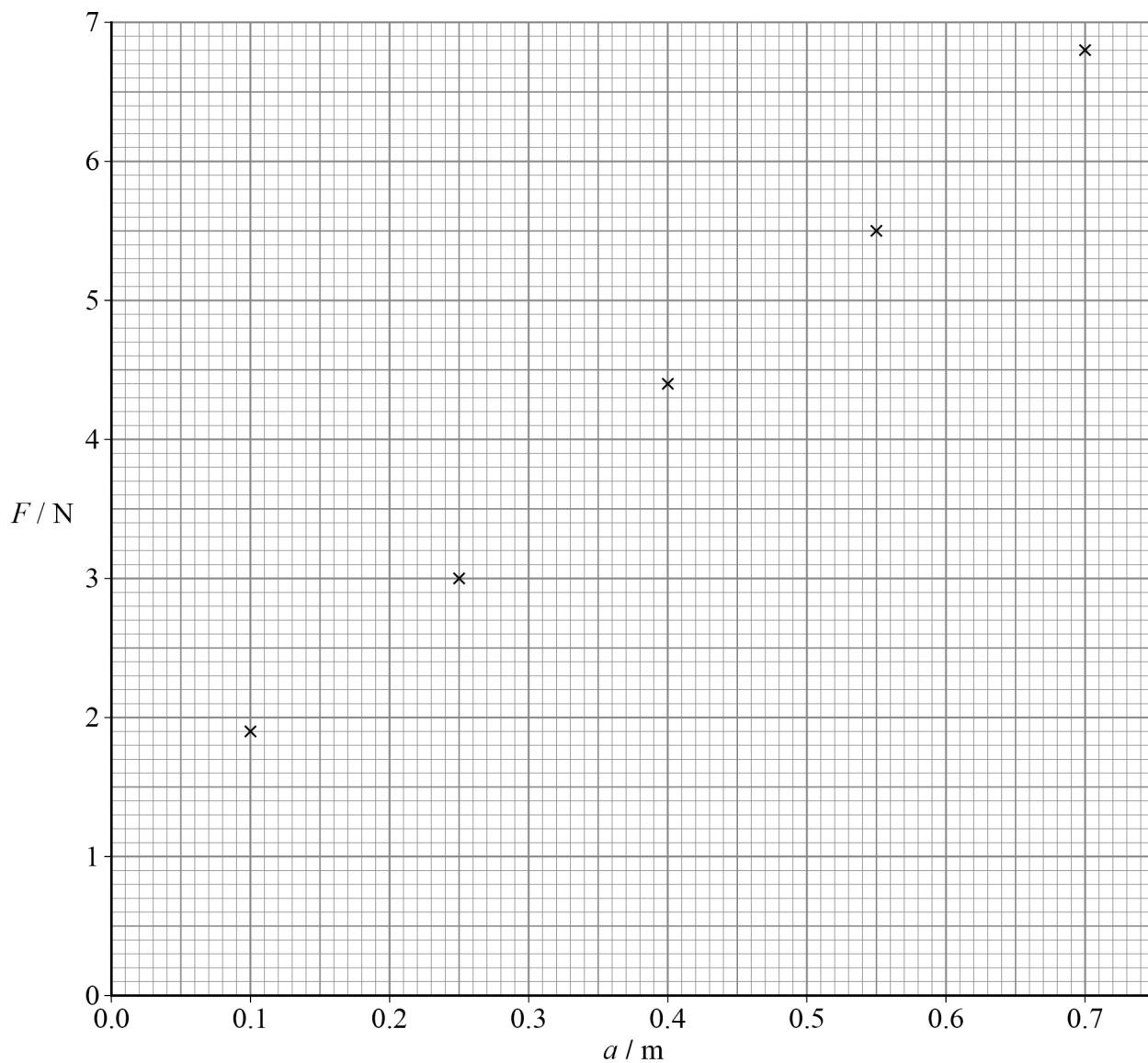
Show, by taking moments about **P**, that

$$F = \frac{Mg}{b}a + \frac{w}{2}$$

[1 mark]

Figure 8 is a plot of F against a .

Figure 8



1 0 . 2 Draw a best-fit line on **Figure 8**.

[1 mark]



10.3 Determine M .

[3 marks]

$M =$ _____ kg

10.4 The force meter has a zero error.

Determine, using **Figure 8**, a value for the zero error.

$w = 2.6 \text{ N}$

[2 marks]

zero error = _____ N

10.5 Explain whether the zero error on the force meter has an effect on the value of M calculated in Question **10.3**.

[1 mark]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

1 3

Do not write
outside the
boxIn which diagram is the moment of the force about **O** the greatest?

[1 mark]

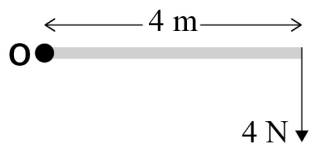
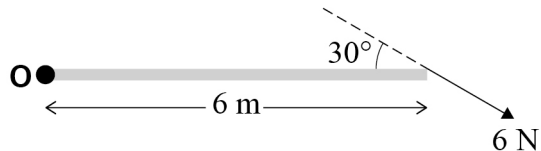
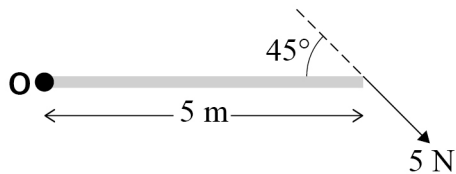
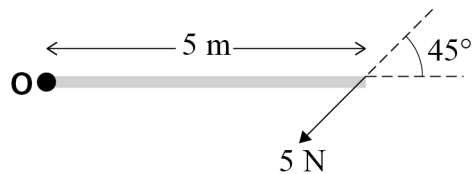
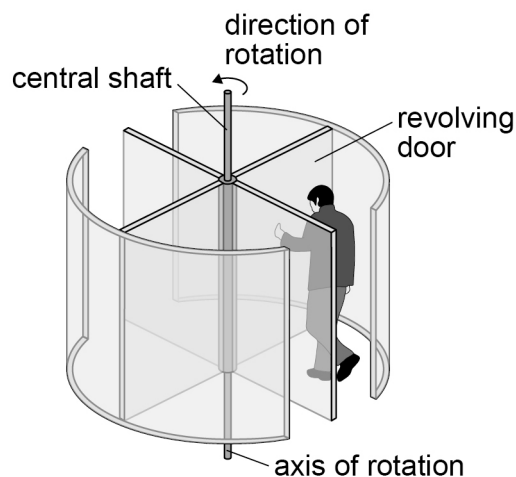
A**B****C****D****A**☐**B**☐**C**☐**D**☐

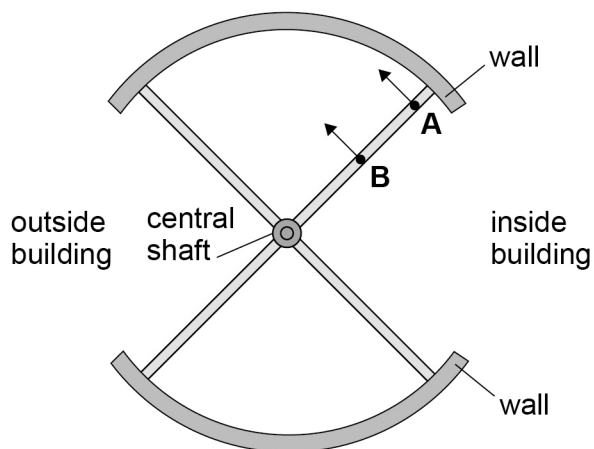
Figure 11 shows a man entering a revolving door, pushing the door to make it turn. The door turns anticlockwise, viewed from above.

Figure 11



side view

Figure 12



view from above

Figure 12 shows two positions **A** and **B** where the door can be pushed to make it turn.

The door does not turn until the applied moment is greater than a particular value.

Compare the minimum force required to turn the door when pushing at **A** with the minimum force required when pushing at **B**.
Explain your answer.

[1 mark]

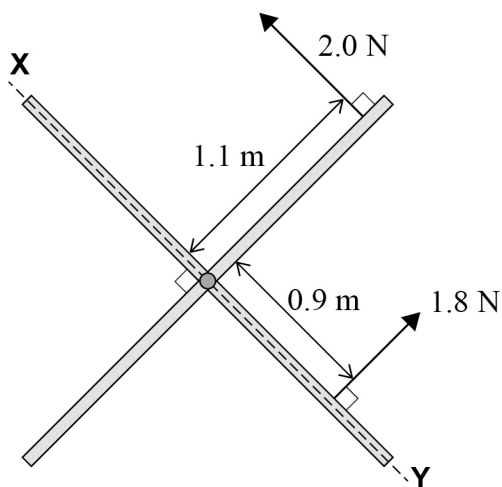
Question 9 continues on the next page



Two people are in different sections of the revolving door. They exert forces on the door, as shown in **Figure 13**. These forces are applied at the same height above the ground.

Figure 13

not to scale



view from above

0 9 . 2

Explain whether the 1.8 N and the 2.0 N forces in **Figure 13** are a couple.

[1 mark]

0 9 . 3

Calculate the resultant moment exerted by the people on the door.

[1 mark]

moment = _____ Nm



The centre of mass of the revolving door is at the axis of rotation, at the central shaft.
The centre of mass of the door does not move.

A reaction force F acts from the shaft to the door.

Determine the magnitude of F and the angle between F and the line **XY** on **Figure 13**.
[3 marks]

magnitude of F = _____ N

angle = _____ °

Draw on **Figure 13** an arrow to represent the direction of F and label the angle
between F and the line **XY**.

[1 mark]



Do not write	Do not write
--------------	--------------

1

1

Which is **not** a unit for energy?

[1 mark]

A eV

☐

B $\text{kg m}^2 \text{s}^{-2}$

☐

C kW h

☐

D N s

☐

Turn over for the next question

Turn over ►



Do not write Do not write

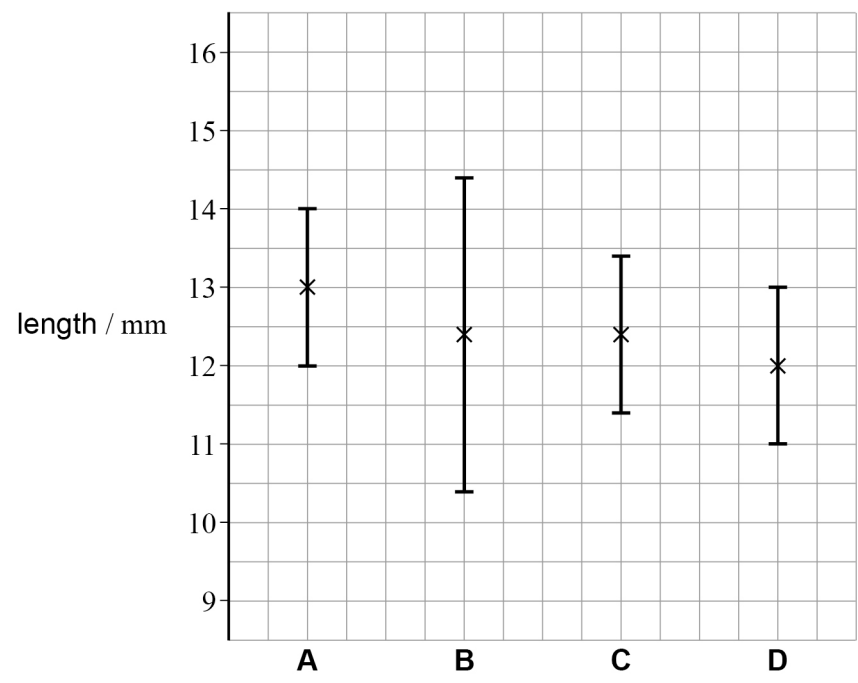
1 2

An object's length is measured.
The five measurements are:

length / mm 10 12 12 14 14

What is the correct way of plotting this length and its error bar?

[1 mark]



- A** ☐
- B** ☐
- C** ☐
- D** ☐

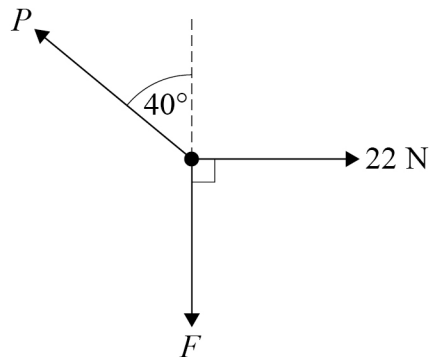


Don't write

1	3
---	---

Do not write
outside the
box

Three coplanar forces P , F and 22 N are in equilibrium.



not to scale

What is the magnitude of F ?

[1 mark]

A 34 N

☐

B 29 N

☐

C 26 N

☐

D 18 N

☐

0 6 . 1

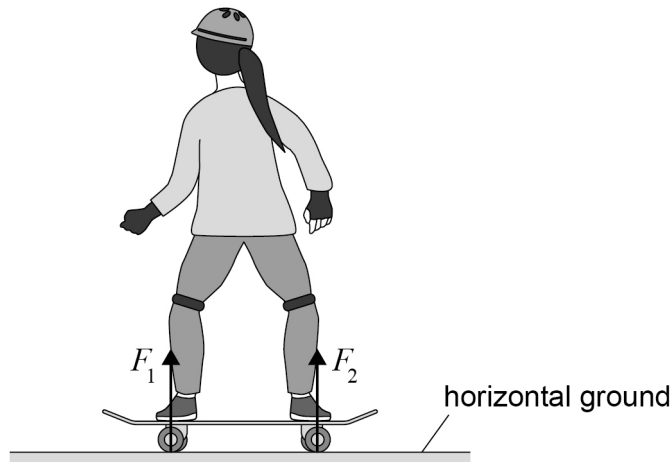
State the **two** conditions for a body to be in equilibrium.

[2 marks]

- 1 _____
- _____
- 2 _____
- _____

Figure 4 shows a child on a skateboard. The skateboard is in equilibrium. The centre of mass of the skateboard is at its midpoint.

Figure 4



0 6 . 2

F_1 and F_2 are the normal reaction forces of the ground acting on the skateboard.

The weight of the child is 360 N.

The mass of the skateboard is 4.8 kg.

The centre of mass of the child is directly above the centre of mass of the skateboard.

Calculate F_1 .

[2 marks]

$$F_1 = \text{_____} \text{ N}$$

Question 6 continues on the next page

Turn over ►



0 6 . 3

The child now changes her position so that the skateboard is balanced on a horizontal bar at **P** as shown in **Figure 5**.

The skateboard is again in equilibrium.

Figure 5

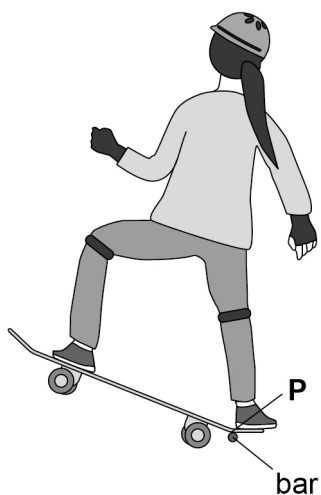
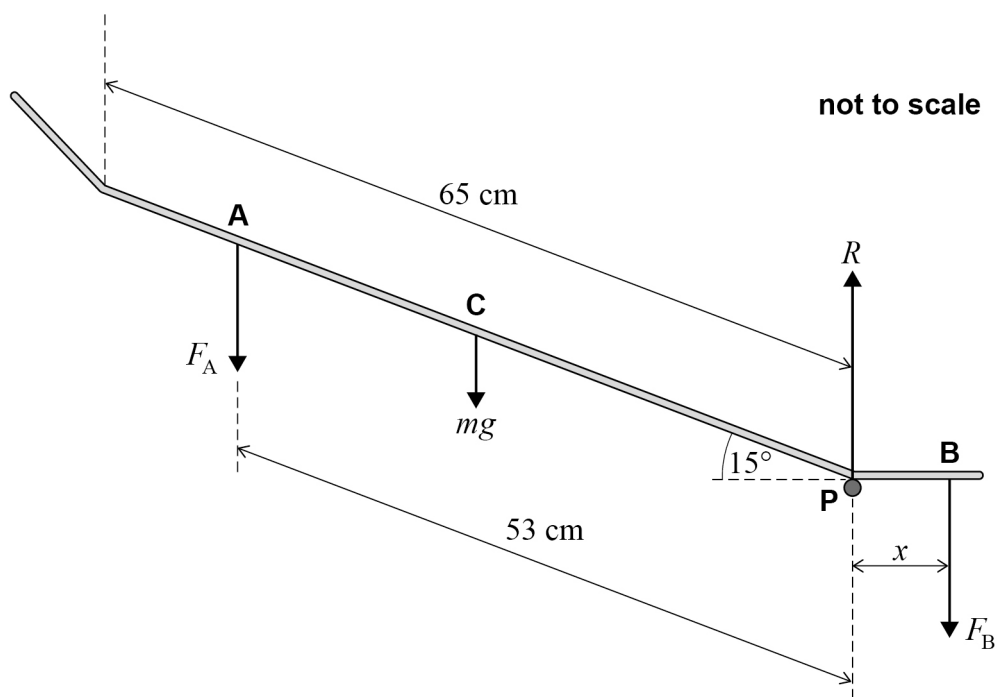


Figure 6 shows the forces acting on the skateboard and the important distances. The bar at **P** exerts a vertical force R on the skateboard. The skateboard is at an angle of 15° to the horizontal.

Figure 6



The child's left foot exerts a vertical force F_A at **A**.

Her right foot exerts a vertical force F_B at **B**.

$$F_B = 11F_A$$

B is a horizontal distance x from **P**.

Calculate x .

[4 marks]

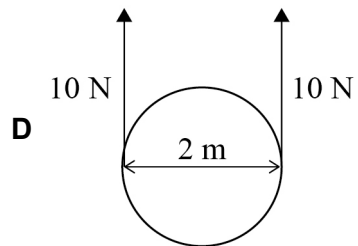
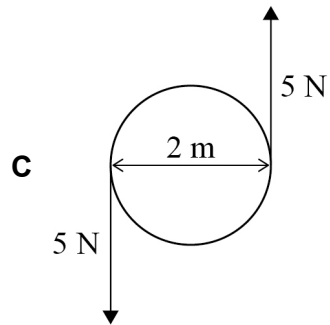
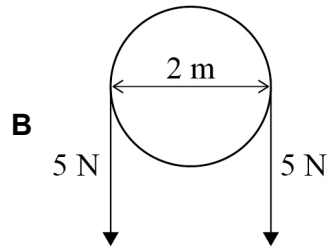
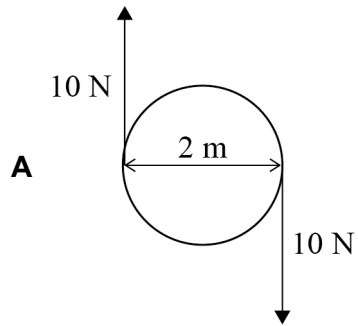
$x =$ _____ cm



Do not write

Which diagram shows a couple with a moment of 20 N m?

[1 mark]



[illegible]

[illegible]

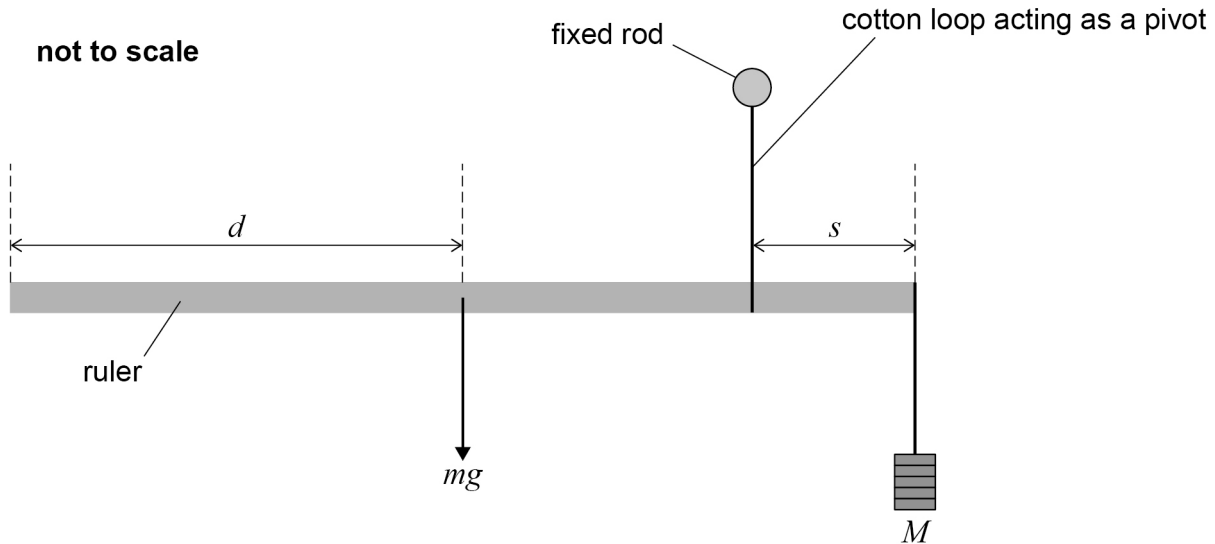
[illegible]

[illegible]

1 0

Figure 10 shows apparatus used to determine the mass m of a uniform ruler. The ruler is supported by a cotton loop that acts as a pivot. A known mass M is hung from one end of the ruler. The position of the ruler is adjusted until it is horizontal and in equilibrium. The horizontal distance between M and the pivot is s . The distance from the end of the ruler to its centre of mass is d .

Figure 10



1 0

. 1

Show that the relationship between M and s is given by

$$M = md \left(\frac{1}{s} \right) - m$$

[1 mark]

Question 10 continues on the next page

Turn over ►



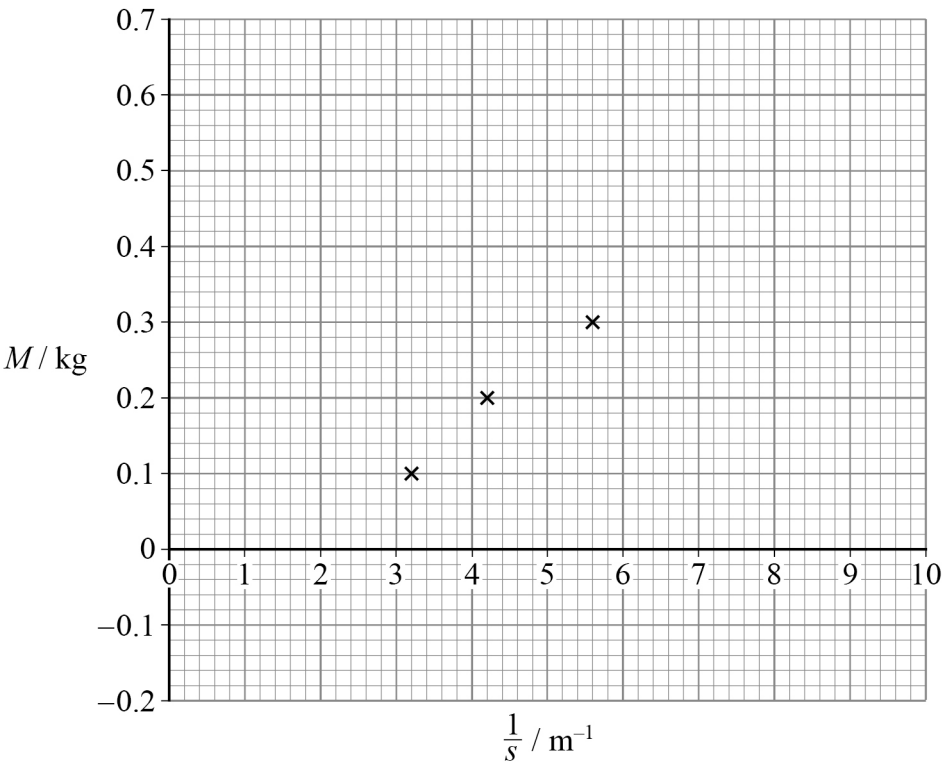
Table 1 shows values of M and $\frac{1}{s}$.

Table 1

M / kg	0.10	0.20	0.30	0.40	0.50	0.60
$\frac{1}{s} / \text{m}^{-1}$	3.22	4.26	5.62	6.71	8.00	9.35

Figure 11 is a plot of M against $\frac{1}{s}$.

Figure 11



1 0 . 2 Plot on **Figure 11** the remaining points from **Table 1**.

[1 mark]

1 0 . 3 Draw on **Figure 11** the line of best fit.

[1 mark]



1	0
---	---

 .

4

 Determine m .

[1 mark]

$m =$ _____ kg

1	0
---	---

 .

5

 Determine d .

[3 marks]

$d =$ _____ m

7

Turn over for the next question



Don't write

Two forces act on an object to form a couple.

State the conditions necessary for the two forces to form a couple.

[2 marks]



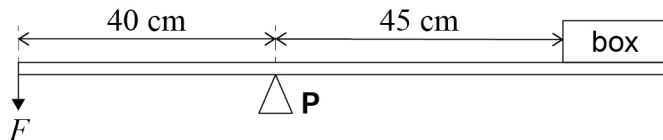
Don't write

1 5

A uniform ruler of mass 75 g and length 100 cm rests on a pivot **P**.

A uniform box of mass 25 g is at one end of the ruler.

A force F acts vertically on the other end of the ruler so that the system is in equilibrium.



What is the magnitude of F ?

[1 mark]

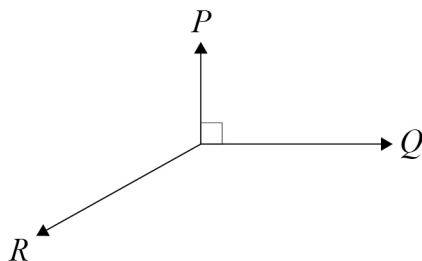
A 0.052 N

B 0.32 N

C 0.46 N

D 0.51 N

An object in equilibrium is acted on by three coplanar forces P , Q and R as shown.



Which row gives possible magnitudes for P , Q and R ?

[1 mark]

	P	Q	R
A	5 N	6 N	7 N
B	8 N	10 N	18 N
C	12 N	16 N	20 N
D	20 N	25 N	34 N

☐
☐
☐
☐


Don't write

0 2 . 1

State what is meant by a vector quantity.

[1 mark]

0 2 . 2

State **one** example of a vector quantity and **one** example of a scalar quantity.

[1 mark]

vector quantity: _____

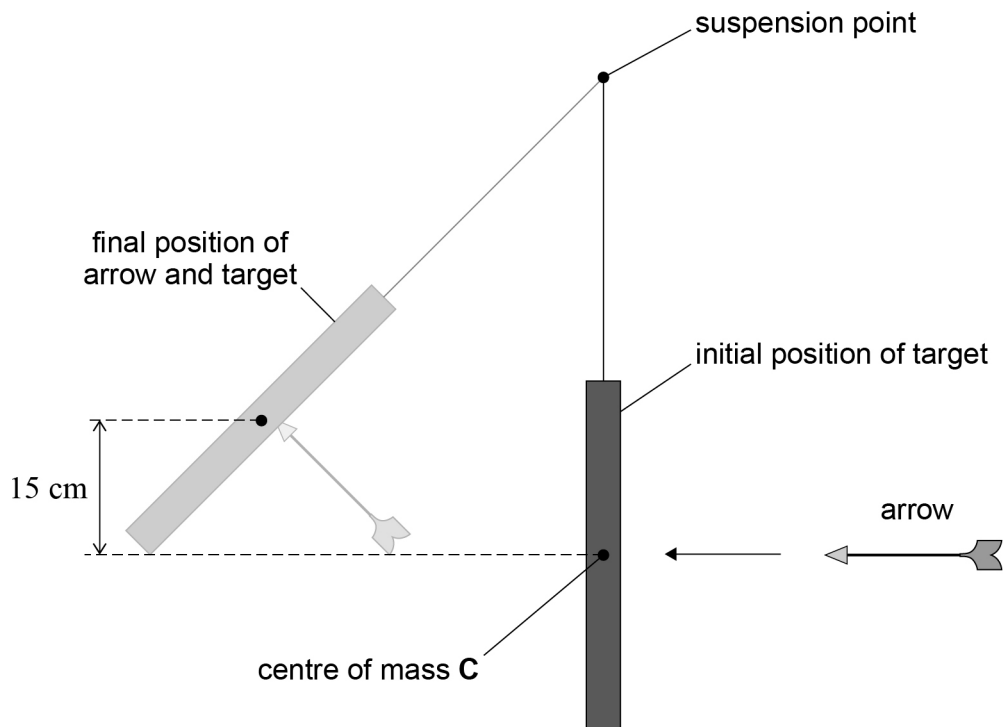
scalar quantity: _____

A target is suspended by a string. The string does not stretch, and the target can swing freely. The centre of mass **C** of the target is in the middle of the target.

An arrow is fired at the target.

The arrow strikes the target horizontally with a velocity u . The arrow sticks into the centre of the target. The maximum change in height of **C** is 15 cm, as shown in **Figure 4**.

Figure 4



Calculate u .

mass of arrow = 95 g
mass of target = 1.2 kg

[3 marks]

$u =$ _____ m s^{-1}



0 7 . 2

On another occasion the arrow strikes the centre of the target at the same velocity u . This time the arrow does **not** stick into the target but bounces backwards.

Deduce how the maximum change in height of **C** compares with the value of 15 cm shown in **Figure 4**.

Calculations are **not** required.

[3 marks]

Question 7 continues on the next page

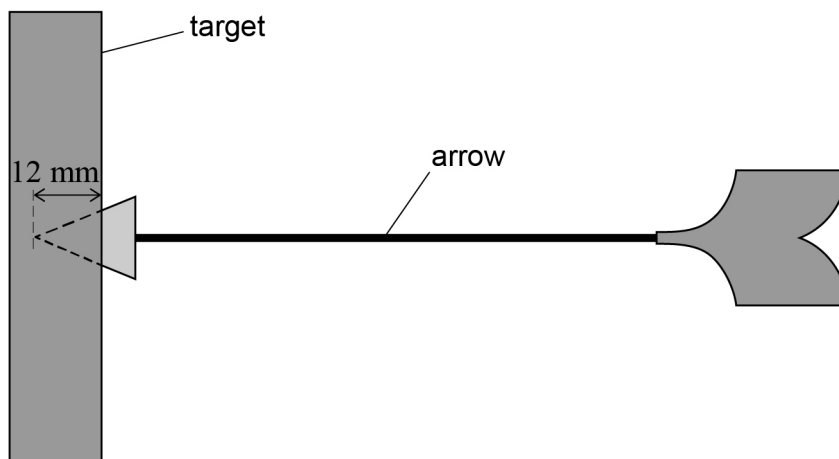


0 7 . 3

The target is now placed on a fixed stand.
The arrow strikes the centre of the target horizontally at a speed of 44 m s^{-1} .
The arrow comes to rest with 12 mm of its length inside the target, as shown in **Figure 5**.

The target does not move.

Figure 5



Calculate the average force that the target applies to the arrow.

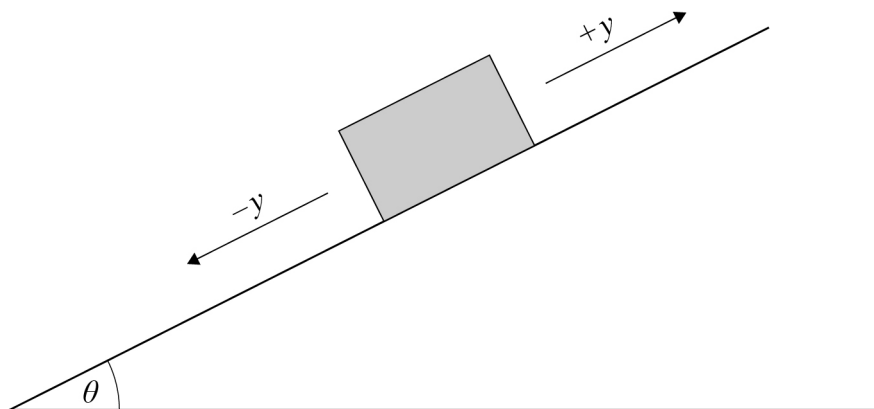
[3 marks]

average force = _____ N

Don't write

2 1

A block of mass m is stationary on a slope.



Which row shows the magnitude and direction of the frictional force acting on the block?

[1 mark]

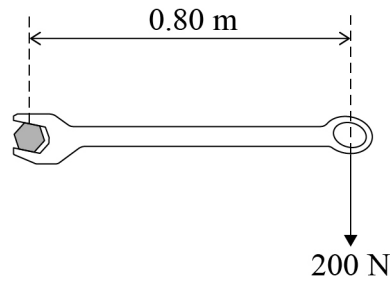
	Magnitude	Direction	
A	$mg \cos \theta$	$+y$	☐
B	$mg \cos \theta$	$-y$	☐
C	$mg \sin \theta$	$+y$	☐
D	$mg \sin \theta$	$-y$	☐



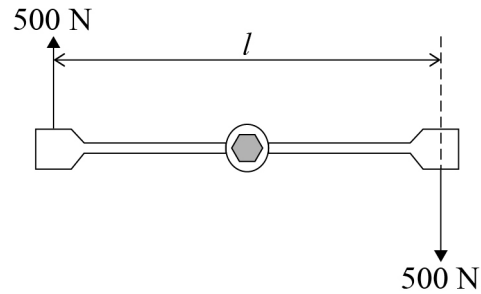
Don't write

2 3

Tool **X** can be used to loosen a bolt by applying a minimum force of 200 N.



Tool **Y** can be used to loosen the same bolt by applying two forces a perpendicular distance l apart. The minimum value of each force is 500 N.



What is l ?

[1 mark]

- A** 0.16 m ☐
- B** 0.32 m ☐
- C** 0.48 m ☐
- D** 0.64 m ☐



Don't write